

TMUA EUCLID

Cheat Sheet

WRITTEN BY TMUA EUCLID

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TMUA Exam Overview

TMUA tests fluent mathematics, fast recognition, and clear logic. Use this as a revision checklist: know the facts, spot the structure, then choose efficiently.

Paper	Focus	Format
Paper 1	Applications of Mathematical Knowledge	20 MCQs, 75 minutes
Paper 2	Mathematical Reasoning	20 MCQs, 75 minutes
Both	No calculator; no formula booklet; no negative marking	Average pace: 3 min 45 s per question

If stuck, eliminate by sign, size, parity, domain, units, graph shape, special cases, or boundary cases. Never leave a question blank.

Number, Ratio, Proportion and Units

Number skills

- Prime factorisation gives factors, multiples, HCF and LCM. For positive integers, $ab = \text{HCF}(a, b) \text{LCM}(a, b)$.
- Standard form: $a \times 10^n$, where $1 \leq a < 10$ and $n \in \mathbb{Z}$.
- Recurring decimals: set $x = \text{decimal}$, multiply by 10, 100, ..., subtract, then solve.
- Bounds: if $x = 4.7$ to 1 d.p., then $4.65 \leq x < 4.75$.
- Exact calculation: keep fractions, surds and π until the end; estimate to check reasonableness.

Ratio, percentages and proportion

$$\text{percentage multiplier} = 1 \pm \frac{p}{100}, \quad A = P(1 + r)^t, \quad \text{simple interest } A = P(1 + rt).$$

Direct proportion: $y = kx^n$. Inverse proportion: $y = k/x^n$. Original value = final value $\div$ multiplier.

Units and compound units

$$\text{speed} = \frac{\text{distance}}{\text{time}}, \quad \text{density} = \frac{\text{mass}}{\text{volume}}, \quad \text{pressure} = \frac{\text{force}}{\text{area}}.$$

Unit conversion: convert units first. Length scale factor k gives area scale factor k^2 and volume scale factor k^3 .

Algebra, Exponentials and Logarithms

Core algebra

$$(a \pm b)^2 = a^2 \pm 2ab + b^2, \quad a^2 - b^2 = (a - b)(a + b), \quad \frac{x + 2}{x} = 1 + \frac{2}{x}.$$

Factor before expanding. For inequalities, move everything to one side, factor, mark critical values, then test intervals.

Indices, surds, exponentials and logs

$$a^m a^n = a^{m+n}, \quad \frac{a^m}{a^n} = a^{m-n}, \quad (a^m)^n = a^{mn}, \quad a^{-n} = \frac{1}{a^n}, \quad a^{1/n} = \sqrt[n]{a}.$$

$$a^x = y \iff \log_a y = x, \quad \log_a(xy) = \log_a x + \log_a y, \quad \log_a\left(\frac{x}{y}\right) = \log_a x - \log_a y, \quad \log_a(x^n) = n \log_a x.$$

Exponential growth/decay: $y = ab^x$; $b > 1$ grows, $0 < b < 1$ decays. Check log arguments are positive.

Quadratics and simultaneous equations

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}, \quad \Delta = b^2 - 4ac, \quad \text{vertex } x = -\frac{b}{2a}.$$

$\Delta > 0$: two roots; $\Delta = 0$: repeated root; $\Delta < 0$: no real roots.

Linear + quadratic simultaneous equations: rearrange the line, substitute into the quadratic, solve, then substitute back.

Division, Factor Theorem, Remainder Theorem and binomial

Remainder Theorem: dividing $f(x)$ by $(x - a)$ leaves remainder $f(a)$. Factor Theorem: $(x - a)$ is a factor $\iff f(a) = 0$.

$$(a + b)^n = \sum_{r=0}^n \binom{n}{r} a^{n-r} b^r, \quad \binom{n}{r} = \frac{n!}{r!(n-r)!}.$$

Use long division or coefficient comparison for algebraic division. For $(a - b)^n$, signs alternate.

Functions, Graphs and Sequences

Functions and transformations

- Composition: $f(g(x))$. Inverse: solve $y = f(x)$ for x , then swap x, y .
- One-to-one: different inputs give different outputs; inverse is valid. Many-to-one: restrict domain first.
- $f(x) + a$: up a ; $f(x - a)$: right a ; $af(x)$: vertical scale a ; $f(ax)$: horizontal scale $1/a$.
- Reflections: $-f(x)$ in x -axis, $f(-x)$ in y -axis.

Common graph shapes

Graph	Remember
Linear	$y = mx + c$; gradient m , intercept c
Quadratic	Parabola; symmetry line through vertex
Cubic	Opposite end behaviour; up to two turning points
Reciprocal	$y = a/x$; asymptotes $x = 0, y = 0$
Exponential	$y = ab^x$; horizontal asymptote often $y = 0$
Logarithmic	$y = \log_a x$; domain $x > 0$; inverse of a^x
Square root	$y = \sqrt{x}$; starts at $(0, 0)$, $x \geq 0$
Modulus	$y = f(x) $; negative parts reflect above the x -axis
Trig	Periodic; check amplitude, period, symmetry and intercepts

Sequences

Arithmetic: $u_n = a + (n - 1)d$, $S_n = \frac{n}{2}(2a + (n - 1)d)$.

$$\text{Geometric: } u_n = ar^{n-1}, \quad S_n = \frac{a(1 - r^n)}{1 - r}, \quad S_\infty = \frac{a}{1 - r} \quad (|r| < 1).$$

Recurrence: use the given start value and iterate carefully; check whether indexing starts at 0 or 1.

Coordinate Geometry and Circles

Lines and circle equations

$$m = \frac{y_2 - y_1}{x_2 - x_1}, \quad y - y_1 = m(x - x_1), \quad d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}.$$

Perpendicular gradients multiply to -1 . Vertical line: $x = c$.

$$(x - a)^2 + (y - b)^2 = r^2.$$

$$x^2 + y^2 + 2gx + 2fy + c = 0, \quad \text{centre } (-g, -f), \quad r = \sqrt{g^2 + f^2 - c}.$$

Complete the square to identify centre and radius.

Circle theorems

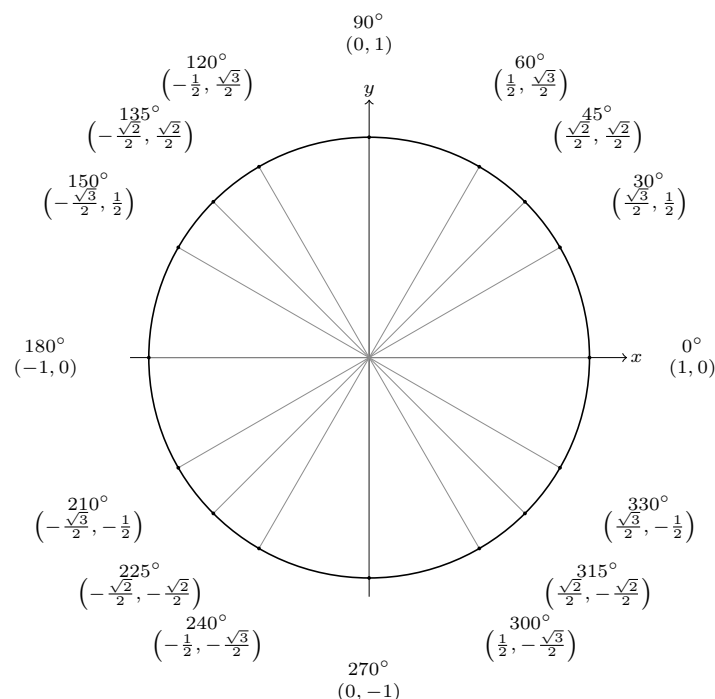
- Perpendicular from centre to chord bisects the chord.
- Angle at centre = $2 \times$ angle at circumference.
- Angles in the same segment are equal; angle in a semicircle is 90° .
- Opposite angles in a cyclic quadrilateral sum to 180° .
- Radius is perpendicular to tangent; tangent-chord angle equals angle in alternate segment.

Trigonometry Essentials

Unit circle and exact values

θ	0°	30°	45°	60°	90°
$\sin \theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\tan \theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	undefined

Unit circle point: $(\cos \theta, \sin \theta)$. Signs: QI (+, +), QII (−, +), QIII (−, −), QIV (+, −).



Triangles, radians and trig equations

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}, \quad a^2 = b^2 + c^2 - 2bc \cos A, \quad \text{Area} = \frac{1}{2}ab \sin C.$$

Ambiguous sine rule: SSA can give B and $180^\circ - B$; check both.

$$180^\circ = \pi \text{ radians}, \quad s = r\theta, \quad \text{sector area} = \frac{1}{2}r^2\theta.$$

Segment area = sector area – triangle area. Use radians for arc/sector formulae.

$$\sin^2 x + \cos^2 x = 1, \quad \tan x = \frac{\sin x}{\cos x}.$$

Periods: sine/cosine 360° or 2π ; tangent 180° or π . For intervals, find all solutions using symmetry and periodicity.

Calculus**Differentiation**

$$\frac{d}{dx}x^n = nx^{n-1}.$$

Tangent: differentiate then substitute point. Normal: negative reciprocal gradient. Stationary points: $dy/dx = 0$.

- Increasing: $dy/dx > 0$; decreasing: $dy/dx < 0$.
- Second derivative: $d^2y/dx^2 > 0$ minimum/concave up; $d^2y/dx^2 < 0$ maximum/concave down.

Integration

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1), \quad \int_a^b f(x) dx = F(b) - F(a).$$

Definite integrals give signed area, which can be negative. Total area: split at roots/intersections and add absolute areas.

$$\text{Trapezium rule} \approx \frac{h}{2}(y_0 + 2y_1 + 2y_2 + \cdots + 2y_{n-1} + y_n).$$

Simple differential equation $dy/dx = f(x)$: integrate, then use the condition to find C .

Geometry, Mensuration and Vectors**Geometry facts**

- Straight line 180° ; around a point 360° ; vertically opposite angles equal.
- Parallel lines: corresponding and alternate angles equal; co-interior angles sum to 180° .
- Polygon interior sum $(n-2)180^\circ$; exterior angles sum 360° .
- Congruence: SSS, SAS, ASA, RHS. Similarity: equal angles and proportional sides.
- Transformations: translation, reflection, rotation, enlargement. Bearings: clockwise from north, three figures.
- Pythagoras: $a^2 + b^2 = c^2$; in 3D, find a face diagonal first. Plans/elevations are 2D views.

Mensuration

$$C = 2\pi r, \quad A_{\text{circle}} = \pi r^2, \quad V_{\text{cylinder}} = \pi r^2 h, \quad V_{\text{prism}} = \text{cross-section area} \times \text{length}.$$

Surface area = exposed faces. Composite shapes: split or subtract.

Vectors

$$|\mathbf{a}| = \sqrt{a_x^2 + a_y^2}, \quad \text{midpoint of } \mathbf{a}, \mathbf{b} = \frac{\mathbf{a} + \mathbf{b}}{2}.$$

Vector proofs: express routes in basis vectors; parallel vectors are scalar multiples; collinear points lie on the same vector line.

Statistics and Probability

Statistics

- Mean = total $\div$ number; median = middle; mode = most frequent; range = max – min.
- Grouped mean: use class midpoints. Quartiles split data into quarters; $\text{IQR} = Q_3 - Q_1$.
- Histogram: frequency density = frequency/class width.
- Cumulative frequency: estimate median, quartiles and percentiles.
- Scatter graphs: correlation and line of best fit; avoid unsafe extrapolation.

Probability

$$P(A') = 1 - P(A), \quad P(A \cup B) = P(A) + P(B) - P(A \cap B), \quad P(A | B) = \frac{P(A \cap B)}{P(B)}.$$

Conditional probability uses the formula above. Mutually exclusive: $P(A \cap B) = 0$. Independent: $P(A \cap B) = P(A)P(B)$. Expected frequency = np .

- Tree diagrams: multiply along branches, add separate routes.
- Dependent events: update probabilities after each outcome.
- Venn diagrams: organise overlaps. Possibility spaces: list equally likely outcomes.

Logic and Proof

Logic language

Phrase	Symbol	Meaning
If P , then Q	$P \Rightarrow Q$	P sufficient for Q
P only if Q	$P \Rightarrow Q$	Q necessary for P
P if Q	$Q \Rightarrow P$	Q sufficient for P
P iff Q	$P \Leftrightarrow Q$	Both directions

Contrapositive: $P \Rightarrow Q$ is equivalent to $\neg Q \Rightarrow \neg P$. Converse $Q \Rightarrow P$ is not generally equivalent.

Quantifiers and proof

$$\neg(\forall x, P(x)) \equiv \exists x \text{ such that } \neg P(x), \quad \neg(\exists x, P(x)) \equiv \forall x, \neg P(x).$$

- Direct proof: assumptions $\rightarrow$ definitions $\rightarrow$ result.
- Cases: split into complete cases. Contradiction: assume false, reach impossible statement.
- Counterexample: one valid example disproves a universal claim.
- Proof ordering: identify assumptions first, conclusion last, and dependencies in between.
- Flaws: proving converse, assuming result, dividing by zero, ignoring domain, relying on diagram/examples.